

WELCOME TO
SITS

Intro Into Hardware

Tuesday 8/19



Debrief

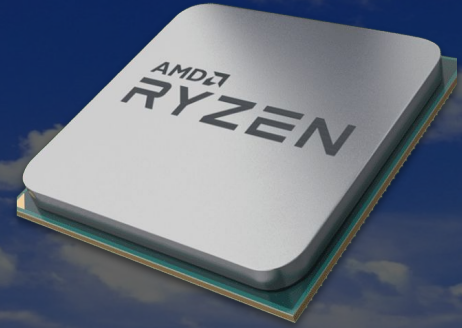
Main goal of this meeting:

- Introduce basic computer hardware (and a bit more)
- Discuss our next meeting
- Answer questions and receive feedback

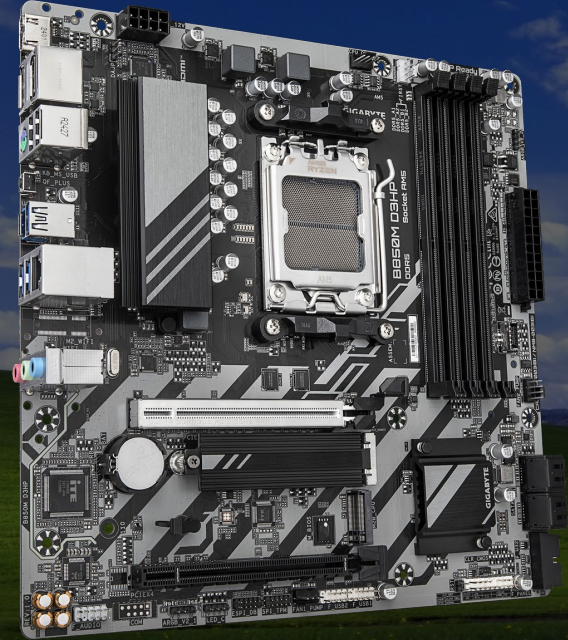


Central Processing Unit

- The “brain” of the computer
- Commands the entire computer
- Executes instructions and calculations
- AMD and Intel are the main 2 producers of desktop CPUs
- Some CPUs are directly connected to the motherboard, such as in laptops



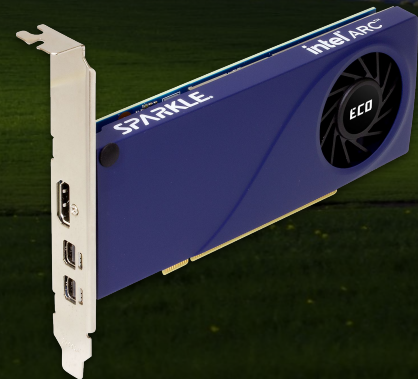
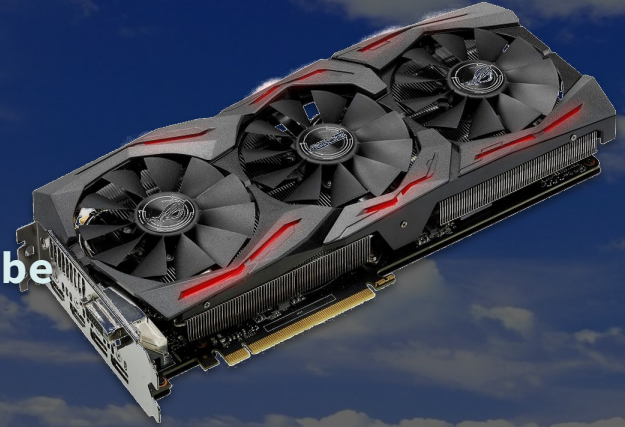
- Physically connects all computer components
- CPUs and some GPUs must be compatible with the motherboards “socket type”
- Some motherboards support more connections / ports than others
- Comes in different sizes (ATX, mATX, mITX)





Graphics Processing Unit

- Handles graphics (visuals)
- Generally used for gaming, but can also be used for other strenuous tasks (3D-Modeling, AI workloads, etc.)
- Nvidia, AMD, and Intel are the main 3 GPU producers though they get shipped and sold by 3rd parties such as AIBs
- Some CPUs are directly connected to the motherboard, such as in laptops





Power Supply Unit

- Powers the entire computer
- Different wattage per unit depending on power requirement
- Comes in fully modular, semi modular, and non-modular cable configurations
- Connects to motherboard, GPU, SSD/HDDs as well as any peripherals like fans





Solid State Drive Hard Disk Drive

- The “memory” of the computer
- Stores all of your files and system data
- SSDs store data with electrically charged “memory cells”
- HDDs store data magnetically on spinning disks





Case / Chassis

- Holds all computer components
- Must be compatible with parts such as the motherboard and power supply
- Main component of good aesthetics
- Some cases come with fans pre-installed





Case Fans CPU Cooler

- Cools the heat generating components
- Generally 3 or more fans should be installed
- CPU coolers are required to keep the CPU from overheating
- GPU fans are pre-installed on the graphics card
- CPU and GPU coolers can also be watercooled





Cables

- All types of cables carry data for different purposes with either electricity or light
- Generally considered a “serial bus”
- Hundreds of different types of cables
- Most cables have a “mini/micro” version of themselves
- Data: USB – Serial
- Video: VGA – DVI – HDMI – DP
- Internet: Ethernet (RJ45) – Fiber





Peripherals

- Allow you to interface with a device
- Also called Human Interface Devices
- Generally consists of computer mice, keyboards, headphones or speakers
- Generally connect over Bluetooth or USB





Extra Information

- Not all computers are stereotypical boxes on a desk. Your smartphone, printer, router and car are all computers on the inside, just different shapes and sizes
- Fun fact: With the correct adapter, you can plug a mouse and keyboard into your phone and control it as if it were a desktop computer



Our Current Mission

Digital Bathroom Pass

Mission goal:

Create a bathroom pass that does not require a piece of paper, pen, etc and is strictly digital

Our Idea:

A scannable QR code near the exit door that brings you to a digital sign in/out page where you can enter your student ID and name. Teachers can export the list of students with all required information

Feel free to speak to an officer if you have any ideas or recommendations!



Next Meeting

Tuesday 8/26

Ground plan:

- Dive deeper into the niche side of computer technology
- Suggestions in purchasing electronics*
- Discuss the current mission





Info

Discord



Website



Instagram

